Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: CRYPTANALYSIS AND CYBER DEFENCE (24CSB3101)

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TITLE OF THE PROJECT	Design and Implementation of an Optimized High-Performance Password Cracking Framework for Cybersecurity Analysis	
S. No.	**University ID**	**Name**
1	2420030091	G.Sirisha
2	2420030264	Y.Yamini
3	2420030136	A.Srinivas
Name of the Guide		**VINISHA VELTHURU**

Abstract

The **Design and Implementation of an Optimized High-Performance Password Cracking Framework for Cybersecurity Analysis** is designed to study password security in a controlled and authorized environment. The framework uses test password hashes to analyze how different password-testing techniques perform. It helps identify weak password patterns and understand the security risks associated with commonly used passwords.

The system supports different approaches such as dictionary-based, brute-force, and rule-based password testing. Each approach can be evaluated based on parameters such as execution time, attempts per second, CPU usage, and memory consumption. This provides a measurable way to compare the effectiveness and performance of different techniques.

The major focus of the project is **optimization**. A basic implementation is first developed and its performance is measured. Optimization techniques such as parallel processing and efficient password-candidate generation are then applied to improve performance and reduce execution time.

Finally, the basic and optimized implementations are compared to calculate the improvement achieved through optimization. The framework can be useful for cybersecurity education, password-security research, and authorized security assessment by helping organizations understand password weaknesses and improve their security mechanisms.

Signature of the Guide HOD